

Kartik Narayan

[LinkedIn](#) | [Github](#) | [Google Scholar](#)
<https://kartik-3004.github.io/portfolio/>

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Summary

I am a 3rd year (final year) Ph.D. student in the Computer Science Department at Johns Hopkins University. My current research interests lie in multimodal LLMs, agents, reasoning and video generation. Looking ahead, I would also love to explore and delve deeper into Embodied AI as a future research direction.

Graduation Date: December 2026

Education

Johns Hopkins University

Ph.D. Computer Science (GPA: 4.0/4.0), Advisor: Dr. Vishal M. Patel

Baltimore, MD

2023 - Present

Indian Institute of Technology Jodhpur

Bachelors in Computer Science and Engineering

Jodhpur, India

2019 - 2023

Research Experience

Salesforce

Ph.D. Research Intern

Advised by Becky Xiangyu Peng, Prafulla Kumar Choubey, Jiaxin Zhang, Silvio Savarese, and Chien-Sheng Wu

Palo Alto, CA

Summer 2026

- Developing efficient multimodal agentic search methods using self-distillation to provide intermediate reward signals.

Apple Inc.

Ph.D. Research Intern

Advised by Lunbo Xu, Xiaohui Tao, and Zhe Gan

Seattle, WA

Spring 2026

- Developed a multi-image, vision-centric benchmark for evaluating multimodal agentic search.
- Analyzed failure modes of multimodal models in agentic search settings involving multiple visual entities.

Apple Inc.

Ph.D. Research Intern

Advised by Chao Jia, Yinfei Yang, and Zhe Gan

Seattle, WA

Summer 2025

- Equipped multimodal LLMs with web-search tools and enabled reasoning over iterative search calls.
- Developed methods to incentivize self-correction and self-verification following an initial model response.

Johns Hopkins University

Research Assistant, Vision and Image Understanding Lab

Advised by Dr. Vishal M. Patel

Baltimore, MD

2023 - Present

- Investigating fundamental challenges in text-to-video (T2V) models, including text-rich video generation.
- Developing multi-agent systems for automated and comprehensive facial analysis.

Indian Institute of Technology Jodhpur

Student Researcher, IAB Lab

Advised by Prof. Richa Singh and Prof. Mayank Vatsa

Jodhpur, India

2022 - 2023

- Studied deepfake video generation under challenging conditions, including multi-face, low-resolution, and occluded videos.
- Introduced deepfake phylogeny by modeling the evolution of deepfakes through sequential face-swapping operations.
- Developed DeSI, a method for identifying deepfake videos on social media and analyzing their propagation.

Publications

1. DeepMMSearch-R1: Empowering Multimodal LLMs in Multimodal Web Search

Kartik Narayan, Yang Xu, Tian Cao, Kavya Nerella, Vishal M. Patel, Navid Shiee, Peter Gräsch, Chao Jia, Yinfei Yang, Zhe Gan

Technical Report. [\[arXiv\]](#)

2. MultiImage-MMSearch: Multi-Image Vision Centric Benchmark for Multimodal Agentic Search in MLLMs

Kartik Narayan, Tian Cao, Zheng Yang, Peng Zhou, Vishal M. Patel, Lunbo Xu, Xiaohui Tao, Zhe Gan

[Under Review.](#)

3. **Efficient Multimodal DeepResearch using Self-Distillation for Intermediate Rewards**
Kartik Narayan, Becky Xiangyu Peng, Prafulla Kumar Choubey, Jiaxin Zhang, Bin Lei,
Vishal M. Patel, Silvio Savarese, Chien-Sheng Wu
[Under Review](#).
4. **Fork Where the Model Changes Its Mind: Belief-Shift Branching for Tree-Structured Reinforcement Learning**
Bin Lei, Yu Li, Prafulla Kumar Choubey, Jiaxin Zhang, Becky Xiangyu Peng, Qinyuan Ye,
Kartik Narayan, Caiwen Ding, Silvio Savarese, Chien-Sheng Wu
[Under Review](#).
5. **FaceMoE: Mixture of Experts for Low-resolution Face Recognition**
Kartik Narayan, Vishal M. Patel
[European Conference on Computer Vision 2026 \(ECCV 2026\)](#)
6. **Silhouette-based Gait Foundation Model**
Dingqiang Ye, Chao Fan, **Kartik Narayan**, Bingzhe Wu, Chengwen Luo, Jianqiang Li, Vishal M. Patel
[European Conference on Computer Vision 2026 \(ECCV 2026 Spotlight\)](#)
7. **RestoreVAR: Visual Autoregressive Generation for All-in-One Image Restoration**
Sudarshan Rajagopalan, **Kartik Narayan**, Vishal M. Patel
[International Conference on Learning Representations 2026 \(ICLR 2026\)](#)
8. **PromptPrint: Behavioral Biometrics Through Natural Language Prompting in LLMs**
Shaiv Patel, **Kartik Narayan**, Vishal M. Patel
[IEEE/IAPR International Joint Conference on Biometrics \(IJCB 2026\)](#)
9. **TransFIRA: Transfer Learning for Face Image Recognizability Assessment**
Allen Tu, **Kartik Narayan**, Joshua Gleason, Jennifer Xu, Matthew Meyn, Tom Goldstein, Vishal M. Patel
[20th IEEE International Conference on Automatic Face and Gesture Recognition \(FG 2026\)](#)
10. **Training-Free Stylized Abstraction**
Aiman Rahman*, **Kartik Narayan***, Vishal M. Patel
[Under Review](#). [[arXiv](#)]
11. **INFER: Implicit Neural Features for Exposing Realism**
Dhananjaya Jayasundara, **Kartik Narayan**, Vishal M. Patel
[Under Review](#). [[paper](#)]
12. **FaceXFormer: A Unified Transformer for Facial Analysis**
Kartik Narayan*, Vibashan VS*, Rama Chellappa, Vishal M. Patel
[Proceedings of the IEEE/CVF International Conference on Computer Vision \(ICCV 2025\)](#)
13. **SegFace: Face Segmentation of Long-Tail Classes**
Kartik Narayan, Vibashan VS, Vishal M. Patel
[Association for the Advancement of Artificial Intelligence \(AAAI 2025\)](#)
14. **PETALface: Parameter Efficient Transfer Learning for Low-resolution Face Recognition**
Kartik Narayan, Nithin Gopalakrishnan Nair, Jennifer Xu, Rama Chellappa, Vishal M. Patel
[IEEE/CVF Winter Conference on Applications of Computer Vision \(WACV 2025 Oral\)](#)
15. **Investigating Social Biases in Multimodal LLMs**
Malsha Perera*, **Kartik Narayan***, Vishal M. Patel
[19th IEEE International Conference on Automatic Face and Gesture Recognition \(FG 2025\)](#)
16. **Improved Representation Learning for Unconstrained Face Recognition**
Nithin Gopalakrishnan Nair*, **Kartik Narayan***, Maitreya Suin, Ram Prabhakar, Jennifer Xu, Soraya Stevens, Joshua Gleason, Nathan Shindman, Rama Chellappa, Vishal M. Patel
[19th IEEE International Conference on Automatic Face and Gesture Recognition \(FG 2025\)](#)
17. **Hyp-OC: Hyperbolic One Class Classification for Face Anti-Spoofing**
Kartik Narayan, Vishal M. Patel
[18th IEEE International Conference on Automatic Face and Gesture Recognition \(FG 2024\)](#)
18. **DF-Platter: Multi-Face Heterogeneous Deepfake Dataset**
Kartik Narayan, Harsh Agarwal, Kartik Thakral, Surbhi Mittal, Mayank Vatsa, Richa Singh
[Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition \(CVPR 2023\)](#)
19. **DeePhyNet: Towards Detecting Phylogeny in Deepfakes**
Kartik Thakral, Harsh Agarwal, **Kartik Narayan**, Surbhi Mittal, Mayank Vatsa, Richa Singh
[IEEE Transactions on Biometrics, Behavior, and Identity Science \(T-BIOM\)](#)
20. **DeePhy: On Deepfake Phylogeny**
Kartik Narayan, Harsh Agarwal, Kartik Thakral, Surbhi Mittal, Mayank Vatsa, Richa Singh

21. **DeSI: Deepfake Source Identifier for Social Media**

Kartik Narayan, Harsh Agarwal, Surbhi Mittal, Kartik Thakral, Suman Kundu, Mayank Vatsa, Richa Singh
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW 2022)

22. **Using Epidemic Modeling, Machine Learning & Control Feedback Strategy for Policy Management of COVID-19**

Kartik Narayan, Heena Rathore, Faycal Znidi
IEEE Access 10 (2022): 98244–98258

23. **Leveraging Ambient Sensing for the Estimation of Curiosity-Driven Human Crowd**

Anirban Das, Kartik Narayan, Suchetana Chakraborty
IEEE International Systems Conference (SysCon 2022)

Academic Services

- **Invited Reviewer:** CVPR, ICCV, ECCV, AAAI, NeurIPS, ICLR, ICML, TPAMI, WaCV, IJCB, TIFS, TBIOM, IJCV, PR

Skills Summary

- **Programming:** Python, C/C++
- **ML/DL:** PyTorch, TensorFlow, Keras, OpenCV, Sklearn, Numpy, Pandas, Matplotlib
- **Development:** HTML, CSS, React.js, Node.js, Bootstrap, Firebase, MongoDB, ReactNative
- **Tools:** Docker, Kubernetes, GIT, AWS

Teaching Experience

Teaching Assistantship: Assisted in teaching by conducting weekly lab sessions, holding special doubt-clearing sessions, preparing quizzes, and grading assignments for the following courses:

- Deep Learning [Spring 2023]
- Introduction to Machine Learning [Fall 2022]
- Pattern Recognition and Machine Learning [Spring 2022]

Co-curricular Activities

- **Internship Head, CSE at the placement cell of IIT Jodhpur** [2021 - 2022]
Coordinated with companies to invite them for campus internships and placements.
- **Head of technical events in Prometeo, IIT Jodhpur** [2021 - 2022]
Led a team of 50+ students to organize technical competitions, attracting over 1,500 participants from across India.
- **Secretary of PHEME, The IITJ Newsletter Club** [2020 - 2021]
Managed a club of 80+ members responsible for creating newsletters, reports, conducting surveys, etc.
- **Student Guide at Student Wellbeing Committee, IITJ** [2020 - 2023]
Mentored 10 freshmen from diverse backgrounds to support their transition into college, guiding them in both academic and non-academic pursuits.